

Particle shape effects on the effective permittivity of anisotropic or isotropic media consisting of aligned or randomly oriented ellipsoidal particles

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Abstract. The effective permittivity (dielectric constant) of anisotropic or isotropic porous media is affected by the shape of particles composing the mixture. Directional permittivities are influenced by extreme aspect ratio particles, often found aligned with the bedding plane of rock or soil. Our objectives were to determine the effects of particle shape and preferential orientation on the effective permittivity of porous media. Confocal spheroids (ellipsoids of revolution) were used to mathematically describe a range of particle shapes from disks to spheres to needles. Dielectric mixing models which account for the polarization due to inclusion shape and axial alignment were used to estimate the shape effect. Permittivity measurements in an anisotropic packing of disk-shaped mica particles using time domain reflectometry showed an alteration of the permittivity due to the shape effect. Two- and three-phase predictions based on *Maxwell-Garnett* [1904] showed trends similar to measurements in anisotropic packings of mica. Particle shape effects can be a significant factor in dielectric permittivity measurements and should be a consideration especially where particle aspect ratio deviates by more than an order of magnitude from that of a sphere (unity). As the particle shape is less spherical, the resulting effective permittivity of the mixture is more similar to the inclusion permittivity and differs more from the permittivity of the background. Ellipsoid size and surface area provide an estimate of the combined effects of bound water and particle shape on the effective mixture permittivity. For high aspect ratio particles, shape effects on the effective permittivity appear to be comparable in magnitude to those of bound water prevalent in clay-sized media.

1. Introduction

Electromagnetic determination of volumetric water content of porous materials such as soil depends upon its constituent components consisting of solids of various size, shape, and mineralogy and of pores filled by air, water, or nonaqueous phase liquids. Modern instruments based on time domain reflectometry (TDR), capacitance sensing, electromagnetic induction, and remotely sensed passive microwave and radar methods facilitate either faster or more accurate moisture content measurements [White and Ziegler, 1995]. These techniques rely on determination of the effective permittivity of the porous medium, and therefore an understanding of the relationship between the effective permittivity and the volumetric water content is of considerable interest. Many factors are known to affect the effective permittivity of soils and various porous media, including water content and status (i.e., bound and free), porosity [Malicki et al., 1996; Friedman, 1997], wetting history [Knight and Nur, 1987], electromagnetic measurement frequency [Campbell, 1990; Heimovaara, 1996], temperature [Or and Wraith, 1999], salinity [Persson and Berndtsson, 1998], and particle or pore shape. Natural soils are character-

ized by a varying degree of anisotropy, which may be due both to soil structure and texture [Arulanandan, 1991].

Particle shape effects have been observed using electromagnetic measurement techniques within various materials. Sillers [1936] found that a minute amount of conducting needle-shaped (prolate) inclusions in wax could produce a serious conductive loss at low frequencies while the same volume of spheres would produce a negligible effect. Kelly et al. [1953] measured higher permittivities of Al flakes (disks) in paraffin than in similar volumetric fractions of spherically shaped Zn and Cu particles. Kraszewski and Nelson [1996] observed shape effects of peanuts on microwave measurements in microwave resonant cavities. The “platy” particle effect due to oblate-shaped clayey particles has been addressed by many researchers measuring the dielectric permittivity of saturated rock [Sen, 1981; Mendelson and Cohen, 1982; Kenyon, 1984; Tyc et al., 1988]. For most soils and rocks having disk-like or clay particles, the most likely situation is probably weak anisotropy, resulting from partial preferential orientation.

Although shape effects have been observed and mathematical theories exist for modeling their effects, little work has been done to verify the orientational and particle shape effects on permittivity measurements in unsaturated soils. Furthermore, confounding effects of bound water and interfacial polarization phenomena have not been separated from shape effects, which would be of great benefit in understanding the role each factor plays. The three-phase geometrical model of Feng and Sen [1985] suggests that low-frequency (MHz) per-

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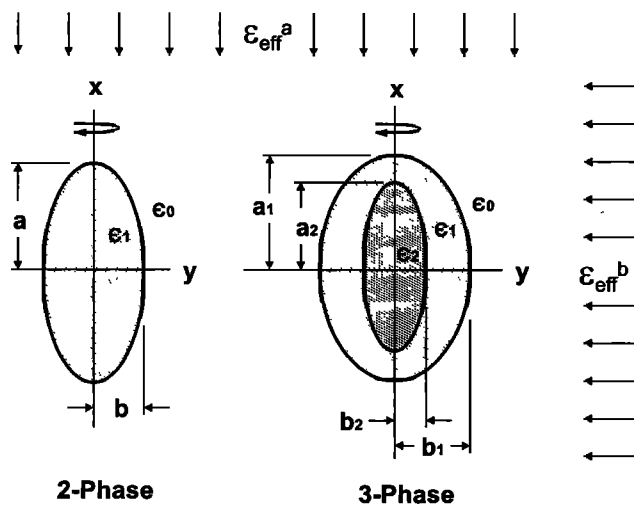


Figure 1. Two- and three-phase systems of spheroidal inclusions of permittivity, ϵ_1 and ϵ_1 and ϵ_2 , respectively, lying in a background of permittivity ϵ_0 . The effective permittivity is referenced to the electrical field alignment, being either parallel to the “a” axis (rotation axis), ϵ_{eff}^a , or to the “b” axes (common axes), ϵ_{eff}^b .

mittivities are very sensitive to grain shape, while permittivities at higher frequencies (GHz) are independent of particle shape. Thevanayagam [1993, 1995] coupled particle shape effects with frequency-domain relaxation effects on permittivity and conductivity in a two-phase mixing theory which matched measurements within the frequency range of 1 to 100 MHz. His model suggests that at high frequencies, particle anisotropy effects on permittivity become negligible, but no data were provided to support this suggestion. Separation of the Maxwell-Wagner interfacial polarization effect from the conductive inclusion geometry effect is conceivable since Maxwell-Wagner effects are proportional to the ionic strength of the solution and occur at lower frequencies. In this work, we only consider the quasi-static polarization due to the geometry of particles and have sought to minimize bound water or double-layer polarization effects in measurements by using larger-diameter particles having a small specific surface area.

Many of the existing dielectric mixing theories assume quasi-static conditions and do not take into account the electromagnetic frequency as a factor affecting the dielectric constant. Such mixing models compute the particle polarization effect on the effective permittivity of mixtures at dc (0 Hz) frequency using an ellipsoidal geometry [Tinga *et al.*, 1973; Sihvola and Lindell, 1990]. The polarization of inclusions within the porous medium arises from the presence of the electric field induced by electromagnetic waves propagating through the medium. The potential range of inclusion shapes considered extends from flat disks to spheres to long needles [Sillers, 1936; Kelly *et al.*, 1953; Doyle and Jacobs, 1992]. Ellipsoidal geometries have been successfully applied in modeling electromagnetic responses in, for example, medical research [Fricke, 1924; Velick and Gorin, 1940]. Various dielectric mixing models and empirical models have been developed, which account for inclusion shape effects for two-phase systems [Polder and van Santen, 1946; de Loor, 1953; Sihvola and Kong, 1988], and concentric spheres or ellipsoids have been used to represent a composite mixture of three or more phases [Tinga *et al.*, 1973; Sihvola and Lindell, 1989; Friedman, 1998]. In the simplest case of spherical

inclusions contained in a background, the effective permittivity is expressed as a function of the inclusion and background permittivities and the volume fraction of inclusions in the mixture.

This work deals with the inclusion (particle) shape effect on the effective permittivity of a saturated/unsaturated porous medium. This effect is especially pronounced where inclusions of extreme aspect ratio are aligned with a common axis forming a bedding plane. Our main objective here was to enhance the understanding of the particle shape effects on the effective permittivity of anisotropic and isotropic porous media consisting of two and three phases by theoretical analysis and measurements in packings of well-defined high aspect ratio particles using TDR.

2. Theoretical Considerations

The particle shape effect on the permittivity of two-phase mixtures can be viewed by considering ellipsoidal inclusions of permittivity ϵ_1 lying in a background of permittivity ϵ_0 (Figure 1). We also consider a three-phase confocal spheroid-background mixture with a two-phase inclusion composed of permittivity ϵ_1 and ϵ_2 again lying in a background of ϵ_0 . All permittivities referred to herein are dimensionless, being expressed with respect to the permittivity of free space ($8.85 \times 10^{-12} \text{ F m}^{-1}$). A uniform external electric field E_e is aligned with one of the three principal axes of the spheroids. For the purpose of simplification all inclusions are spheroids (ellipsoids of revolution) having a unique axial dimension “a” with two common axial dimensions “b” and “c” (i.e., $b = c$). Therefore we refer to the effective mixture permittivity ϵ_{eff} with respect to the electrical field alignment parallel to the “a” axis as ϵ_{eff}^a , and, correspondingly, ϵ_{eff}^b refers to an electrical field applied parallel to either “b” or “c” or, for that matter, any field perpendicular to “a.” The largest dimension, a or b, should be smaller than $\lambda_w/2\pi$ to avoid scattering effects, where λ_w is the free-space wavelength of the measurement device [Yaghjian, 1980].

A general ellipsoid having axial dimensions a, b, and c can be described in Cartesian coordinates by a cubic equation in u (a scalar) as

$$\frac{x^2}{u + a^2} + \frac{y^2}{u + b^2} + \frac{z^2}{u + c^2} = 1, \quad (1)$$

where x, y, and z are the position coordinates of any point on the surface of the ellipsoid. Multiple shells of confocal ellipsoids can be described by adjusting the scalar u, where $u < 0$ or $u > 0$ produce internal or external shells, respectively, which are confocal to some reference ellipsoid (i.e., $u = 0$). If we let $a_j^2 = u_j + a^2$, then the volume V_j of the jth ellipsoidal shell is given as

$$V_j = \frac{4}{3} \pi \sqrt{(u_j + a^2)(u_j + b^2)(u_j + c^2)}, \quad (2)$$

where u_j can be computed directly from the known ellipsoid volume V_j and a, b, and c (see appendix A for solution). Furthermore, the confocality requirement of a multilayer ellipsoid dictates that the semi-axes a_j , b_j , and c_j , of each ellipsoidal boundary must satisfy the condition $a_1^2 - a_2^2 = b_1^2 - b_2^2 = c_1^2 - c_2^2$. The axial aspect ratio of the solid phase reference shell is held constant, while the internal/external (e.g., water or air phase) shell aspect ratio varies according to the respective volumetric fraction, which is why the external

spheroid in a confocal set tends to be more spherical (see Figure 1).

The polarization of a particle P , due to an electrical field within a porous medium, increases the interface free-charge density, thereby modifying ϵ_{eff} . The effective permittivity is associated with the electrical flux density (displacement) D , the external electric field E_e , and P according to

$$D = \epsilon_{\text{eff}} E_e + P. \quad (3)$$

The polarization component of this relation is of special interest for anisotropic dielectric mixtures which contain highly polarizable inclusions. The polarization can be expressed as the product of the number of inclusions per unit volume n_v times their respective dipole moment p ($P = n_v p$). The ratio of p to E_e is the polarizability α ($p = \alpha E_e$), which can be used to express the resulting polarization as

$$P = n_v \alpha E_e. \quad (4)$$

Ultimately, the polarizability is a function of the inclusion geometry (independent of volume) and orientation with respect to the applied electrical field, which are both taken into account by what is termed the depolarization factor. The depolarization factor is a descriptor of the extent to which the inclusion polarization is diminished according to the particle's shape and orientation with respect to the applied electrical field. Here we take N^i to be the depolarization factor in the i th axial direction which, for a general ellipsoid, is given as [Landau and Lifshitz, 1960]

$$N^i = \int_0^\infty \frac{(abc) du}{2(u + i^2) \sqrt{(u + a^2)(u + b^2)(u + c^2)}}, \quad i = a, b, c, \quad (5)$$

where $N^a + N^b + N^c = 1$. Although graphical and numerical approximations for N^i are available [Sillers, 1936; Osborn, 1945; Doyle and Jacobs, 1992], they all treat geometries of disks, spheres, and needles separately. A continuous function for N^a was found by numerically integrating (6) and best fitting a second-order equation to the integrated values ($r^2 = 0.9999$). This relation for the special case of a spheroid provides the depolarization factor with respect to the axis of rotation, N^a , as a function of aspect ratio, $a : b$, from which $N^b = N^c$ follows

$$N^a = \frac{1}{1 + 1.6(a:b) + 0.4(a:b)^2} \quad N^b = 0.5(1 - N^a). \quad (6)$$

These relations were plotted (Figure 2) for the range of axial ratios spanning 3 orders of magnitude in each direction from the depolarization factors of a sphere ($N^{a,b,c} = 1/3, 1/3, 1/3$) to the other two oblate and prolate extremes of thin disks ($N^{a,b,c} = 1, 0, 0$) and long needles ($N^{a,b,c} = 0, 0.5, 0.5$). Since the axial ratios of confocal ellipsoids differ (e.g., $a_1 : b_1 \neq a_2 : b_2$), the depolarization factors vary depending on the volume ratio of each set of confocal ellipsoids.

A family of mixing models describes the effective permittivity of a two-phase anisotropic or isotropic mixture of spheroidal inclusions of volume fraction, $\phi_1 = V_1/V_t$, where V_t is the total representative volume or unit cell. For the isotropic case a summation over all three axial dimensions of the particle, a , b , and c , is given in a general form following Sihvola and Kong [1988]

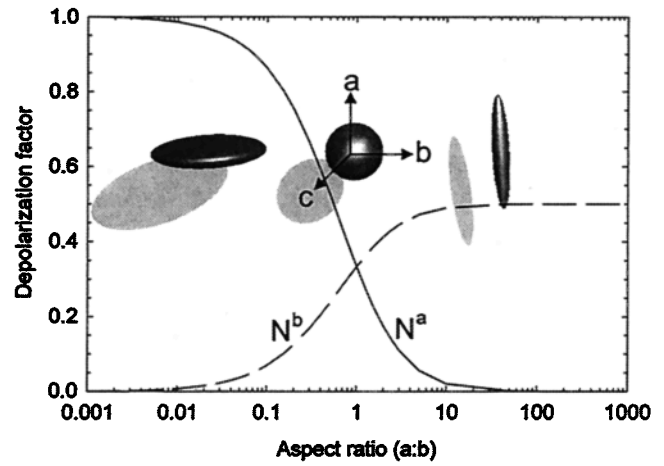


Figure 2. Depolarization factor as a function of aspect ratio and axial-electrical field alignment, where N^a is the depolarization factor for a particle whose rotation axis a is aligned parallel to the electrical field and N^b is the depolarization factor with respect to the common axes.

$$\epsilon_{\text{eff}} = \epsilon_0 + \left\{ \left[\sum_{i=a,b,c} \frac{\phi(\epsilon_1 - \epsilon_0)[\epsilon_0 + \nu(\epsilon_{\text{eff}} - \epsilon_0)]}{3[\epsilon_0 + \nu(\epsilon_{\text{eff}} - \epsilon_0) + N^i(\epsilon_1 - \epsilon_0)]} \right] \cdot \left[1 - \sum_{i=a,b,c} \frac{\phi N^i(\epsilon_1 - \epsilon_0)}{3[\epsilon_0 + \nu(\epsilon_{\text{eff}} - \epsilon_0) + N^i(\epsilon_1 - \epsilon_0)]} \right]^{-1} \right\}. \quad (7)$$

The resulting effective permittivity of the anisotropic mixture in the i th direction is strongly dependent on the particle's aspect ratio and alignment with respect to the applied electric field. For the anisotropic case it is sufficient to carry out the analysis for only the axis of interest (i.e., a , b , or c) written as

$$\epsilon_{\text{eff}}^i = \epsilon_0 + \left\{ \left[\frac{\phi_1[\epsilon_0 + \nu(\epsilon_{\text{eff}}^i - \epsilon_0)](\epsilon_1 - \epsilon_0)}{[\epsilon_0 + \nu(\epsilon_{\text{eff}}^i - \epsilon_0)] + N^i(\epsilon_1 - \epsilon_0)} \right] \cdot \left[1 - \frac{\phi_1 N^i(\epsilon_1 - \epsilon_0)}{[\epsilon_0 + \nu(\epsilon_{\text{eff}}^i - \epsilon_0)] + N^i(\epsilon_1 - \epsilon_0)} \right]^{-1} \right\}, \quad (8)$$

where ν is a parameter which gives the Maxwell-Garnett [1905] (MG) mixing rule when equal to 0, the Polder and van Santen [1946] (PVS) or effective medium approximation mixing formula when $\nu = 1 - N^i$, and the coherent potential (CP) [Tsang et al., 1985] mixing rule when $\nu = 1$. These three classical mean field theories represent a range of different possible approaches for answering the question of how to represent the background permittivity as "felt" by the inclusions within the mixing scheme. The MG solution refers to a dilute inclusion concentration but has also been successfully applied in the past to mixtures having a relatively high inclusion concentration [Sihvola, 1996]. It generally produces an effective permittivity which is closer to the background permittivity than PVS or CP because it neglects the second-order effects due to neighboring inclusions. The CP formula assumes that the apparent background permittivity is equal to the mixture permittivity, and therefore its predicted ϵ_{eff} produces the largest contrast compared to the background permittivity. A compromise between MG and CP is the PVS formula, which treats the inclusions and environment symmetrically leading to an equal influence of each of the two-phase permittivities. A technical advantage of the MG formula is that it results in a explicit

expression for ϵ_{eff} , making it more convenient to use than the implicit PVS and CP formulas.

Modeling the dielectric permittivity of three-phase systems is more difficult because of the complexity of geometry, phase configuration, and distinguishing between host and inclusion. One approach is to allow for multiple independent inclusion geometries, each of different permittivities, contained in a background [de Loor, 1953]. Another approach uses concentric spheres or confocal ellipsoids as inclusions, which represent the different medium phases [Tinga *et al.*, 1973; Sihvola and Lindell, 1989, 1990; Friedman, 1998]. While such mixing theories tend to be numerically more complex, the ability to mathematically include the three or four major phases of soils (solid particles, free water, bound water, and air) is appealing. For the three-phase mixtures we have selected a mixing scheme based on the MG theory for two confocal inclusion phases (ϵ_1 and ϵ_2) contained in a background representing the third phase (ϵ_0). The effective permittivity of an anisotropic or isotropic three-phase confocal system of ellipsoids was derived by Sihvola and Lindell [1990, equation (48)] for any number of confocal ellipsoids, written for the isotropic case as

$$\epsilon_{\text{eff}} = \epsilon_0 + \frac{\frac{\epsilon_0}{3} \sum_{i=a,b,c} (n_v \alpha^i / \epsilon_0)}{1 - \frac{1}{3} \sum_{i=a,b,c} N_1^i (n_v \alpha^i / \epsilon_0)}, \quad (9)$$

and for the anisotropic case the expression is

$$\epsilon_{\text{eff}}^i = \epsilon_0 + \frac{\epsilon_0 (n_v \alpha^i / \epsilon_0)}{1 - N_1^i (n_v \alpha^i / \epsilon_0)}, \quad (10)$$

where the polarizability term in parentheses is given as a series expansion written here for a dual confocal ellipsoid system as

$$\begin{aligned} \frac{n_v \alpha^i}{\epsilon_0} = (\phi_1 + \phi_2) & \left\{ \left[(\epsilon_1 - \epsilon_0) \right. \right. \\ & + \left. \left[\epsilon_1 + N_1^i (\epsilon_0 - \epsilon_1) \right] \frac{(\epsilon_2 - \epsilon_1) \frac{\phi_2}{(\phi_1 + \phi_2)}}{[\epsilon_1 + N_2^i (\epsilon_2 - \epsilon_1)]} \right] \\ & \cdot \left[\left[\epsilon_0 + N_1^i (\epsilon_1 - \epsilon_0) \right] + N_1^i (1 - N_1^i) (\epsilon_1 - \epsilon_0) \right. \\ & \cdot \left. \left. \frac{(\epsilon_2 - \epsilon_1) \frac{\phi_2}{(\phi_1 + \phi_2)}}{[\epsilon_1 + N_2^i (\epsilon_2 - \epsilon_1)]} \right]^{-1} \right\}, \quad (11) \end{aligned}$$

where ϕ_1 and ϕ_2 are the volumetric fractions and N_1^i and N_2^i are the depolarization factors of the outer and inner ellipsoids, respectively. Various combinations of solid, liquid, and gas phases are possible, but the more representative would be a solid surrounded by water films in a background of air for a relatively dry soil. Near saturation the more representative choice of background is water, and we choose the air as the inner inclusion with the solid particle forming the intermediate shell, thus maintaining the solid-water interface as suggested by Friedman [1998].

The effective permittivity for the scenario where the electrical field or potential gradient is directed arbitrarily, forming

angles β^a , β^b , and β^c relative to the a , b , and c coordinate axes of uniformly aligned ellipsoids, respectively, is given by

$$\epsilon_{\text{eff}}^{\beta} = \epsilon_{\text{eff}}^a \cos^2 \beta^a + \epsilon_{\text{eff}}^b \cos^2 \beta^b + \epsilon_{\text{eff}}^c \cos^2 \beta^c, \quad (12)$$

where ϵ_{eff}^a , ϵ_{eff}^b , and ϵ_{eff}^c are the effective permittivities along the principal axes of anisotropy.

Hereafter we refer to the effective permittivity of either perfectly aligned spheroidal particles, forming a uniaxial anisotropic medium described by ϵ_{eff}^a or ϵ_{eff}^b ((8) and (10)), or we refer to completely isotropic packings, designated ϵ_{eff} ((7) and (9)). For reference purposes we will plot the empirical relationship of Topp *et al.* [1980], for the $\epsilon_{\text{eff}}(\theta)$ relationship of isotropic mineral soils, given as

$$\epsilon_{\text{eff}} = 3.03 + 9.3\theta + 146\theta^2 - 76.7\theta^3, \quad (13)$$

where θ is the volumetric water content.

3. Materials and Methods

Particulate media exhibiting relatively narrow distributions of size and shape were obtained for measurement and determination of particle shape effects. Because high-aspect ratio disk-shaped particles produce the most significant alteration of permittivity compared to that of a sphere, mica particles of relatively narrow particle size distribution and high aspect ratio were selected. Phlogopite Mica (Suzorite Mica Products, Boucherville, Quebec, Canada), previously manufacturer-sieved, was rinsed in deionized water and oven dried. The mean (μ) and standard deviations (σ) of the particle diameter ($2b$), thickness ($2a$), and aspect ratio ($a:b$) are given in Table 1. Particle density was determined by the picnometer method. Cross-sectional dimensions and surface area were determined using microscopically scanned images (10 by 10 mm) and image analysis software (Media Cybernetics, L. P., Image Pro Plus version 4.0 for Windows 95/NT/98, Silver Spring, Md., 1998). The square root of two orthogonally measured maximal and minimal diameters was taken as the average diameter μ_{2b} , and the average thickness μ_{2a} was determined from μ_{2b} , the total mass and the particle density of each particle size class. The average aspect ratio $\mu_{a:b}$ for each particle size is given as μ_{2a}/μ_{2b} , and the standard deviation is estimated according to $\sigma_{a:b} = (\sigma_{2a}/\mu_{2a} + \sigma_{2b}/\mu_{2b})\mu_{2a}/\mu_{2b}$. The three different sizes were loosely deposited into separate 5-L beakers filled with saline solutions of 5.2 (0.25 mm), 2.1 (0.5 mm), and 5.1 (1 mm) dS m^{-1} . The saline solutions were used for simultaneous determination of electrical conductivity to be reported elsewhere. Three-wire waveguides made of 4-mm diameter rods of length 100 mm and spaced at 50 mm were prepositioned, two in each beaker, one with rods parallel to the expected horizontal bedding plane of the mica and the other with rods perpendicular to it (Figure 3). The mica was gradually desaturated by applying suction using a water-filled cylindrical ceramic cup located in the bottom of the container. Daily permittivity measurements were recorded using the WINTDR98 waveform analysis software (D. Or *et al.*, WinTDR98 V4.0—User's Guide (Windows-based time domain reflectometry program for measurement of soil water content and electrical conductivity), available at <http://psb.usu.edu/wintdr99/index.html> (posted October 1, 1999 and verified May 3, 2000) 1998) and a Tektronix 1502C cable tester. An average dielectric constant of the phlogopite mica of 6.35

Table 1. Measured and Calculated Parameters of Mica Particles

	Mica Nominal Diameter, mm			Units
	1	0.5	0.25	
Manufacturer designation	20-S	40-S	HK-60	
Manufacturer sieved sizes	0.42–2.0	0.21–0.85	0.08–0.42	mm
Number of particles sampled	37	116	374	
Total group mass	8.55	2.50	1.99	mg
Total cross-sectional area	0.698	0.515	0.581	cm ²
Particle density	2.858	2.876	2.858	g cm ⁻³
μ (mean)—diameter	1510	738	438	μm
σ (standard deviation)—diameter	413	198	120	μm
μ —thickness	58.8	24.4	18.8	μm
σ —thickness	40.5	26.4	44.4	μm
μ —aspect ratio	0.039	0.033	0.043	
σ —aspect ratio	0.037	0.045	0.113	
Specific surface area	163	412	584	cm ² g ⁻¹

(manufacturer reported for 100 Hz) was used in calculations of the effective permittivity.

Soda-lime silicate glass beads (Mo-Sci Corp., Rolla, Missouri, United States of America, and Sigmund Lindner, War-mensteinach, Germany) having a reported dielectric constant of 7.5 at 1 MHz [Bocko and Whitney, 1991, p. 1049] and mean diameters of 500, 250, 100, 50, and 25 μm were also used. Seven-wire TDR probes were constructed having 150-mm-long, 3-mm-diameter outer rods and a 5-mm diameter central rod (12-mm spacing between). TDR measurements were made with the seven-wire coaxial probes in water-saturated glass beads contained within plexiglass boxes measuring 50 $\times$ 50 $\times$ 170 mm internally.

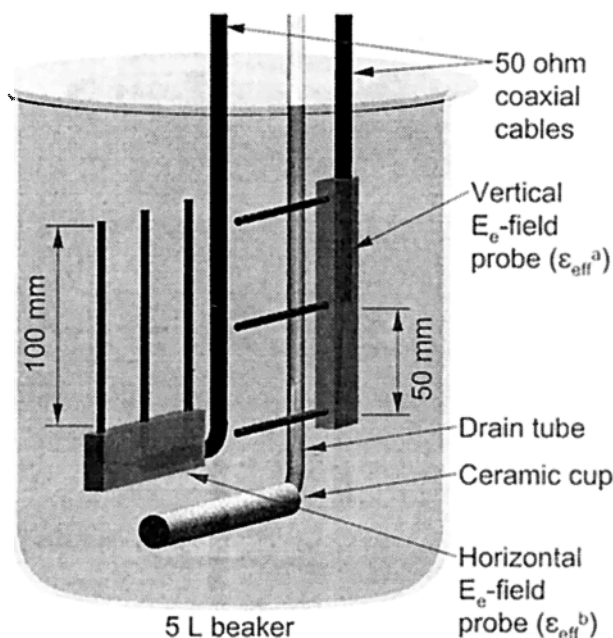


Figure 3. Measurement setup showing probe dimensions and configuration in a 5-L beaker where mica packings were deposited in water and allowed to settle, forming a horizontal bedding plane.

4. Results and Discussion

4.1. Two-Phase Modeling

A two-phase porous medium consisting of a solid inclusion ($\epsilon_1 = 5$) of volume fraction 0.5 and having an aspect ratio of from 10^{-3} to 10^3 lying in a background of either air ($\epsilon_0 = 1$) or water ($\epsilon_0 = 80$) is considered. Equations (7) and (8) give permittivity estimates for the isotropic, ϵ_{eff} , and the anisotropic, ϵ_{eff}^a and ϵ_{eff}^b , for the solid-air mixture denoted SA in Figure 4a. However, we found CP incompatible for the large dielectric contrast of a solid-water (SW) mixture ($\epsilon_0/\epsilon_1 = 16$) which is likened to a saturated soil (Figure 4b). Arithmetic means of the phase permittivities ($\epsilon_{\text{SA}} = 3$ and $\epsilon_{\text{SW}} = 42.5$) result from MG and PVS models for the ϵ_{eff}^b of “infinitely” thin disks ($a:b = 10^{-3}$) and for the ϵ_{eff}^a of “infinitely” long needles ($a:b = 10^3$). Harmonic mean permittivities result for MG and PVS models for ϵ_{eff}^a of thin disks ($\epsilon_{\text{SA}} = 1.67$ in Figure 4a and $\epsilon_{\text{SW}} = 9.41$ in Figure 4b). The MG model consistently predicts an effective permittivity closer to the background permittivity when compared to other mixing models. Needle-shaped particles produce unique permittivities perpendicular to their long axis (ϵ_{eff}^b), which case converges to a two-dimensional circular problem. Considering the effect of particle shape, with a sphere as reference, needles enhance ϵ_{eff}^a measured along their long axis, because of the dipole resulting from their large length, and disks produce much lower permittivities perpendicular to their face (ϵ_{eff}^a) because of their large depolarizing influence (i.e., $N^a = 1$). In accordance, disks enhance ϵ_{eff}^b more than ϵ_{eff}^b of spheres or needles because of the polarization resulting from the large disk diameter. Realistically, disks (and needles) are not only difficult to randomly arrange but would pose much larger porosities than spherical particles, resulting in mixture permittivities closer to that of the background (see discussion of particle shape effect on porosity which follows). Overall, the majority of the changes due to particle shape effects occur within an order of magnitude of the aspect ratio of a sphere ($a:b = 1$), beyond which the shape effect on the effective permittivity changes little. Particle geometrical effects on predicted permittivities are therefore of greatest influence where $a:b < 0.1$ or $a:b > 10$. It is also worth noting that the Topp *et al.* [1980] empirical relationship for $\theta = 0.5$ gives a value (34.6) similar to the MG prediction for spheres (35.6). The modeled isotropic permittivity predictions

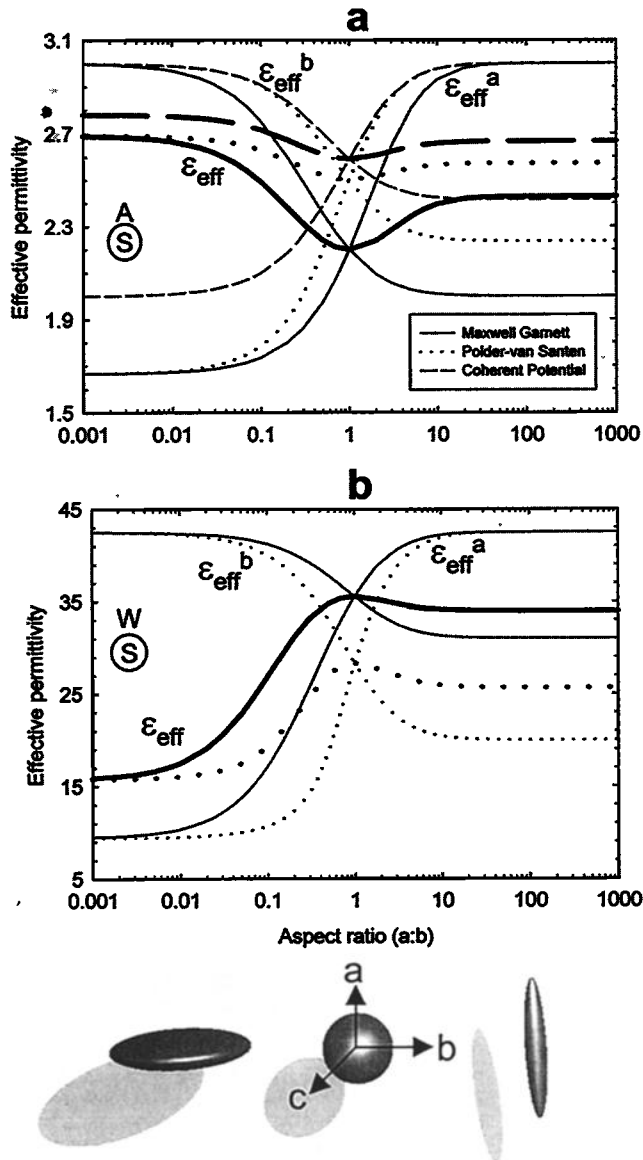


Figure 4. Effective permittivity as a function of particle aspect ratio for two-phase mixtures of (a) solids in air and (b) solids in water, both having porosities of 0.5. Mixing models are according to line type with permittivities of aligned particles as indicated by ϵ_{eff}^a and ϵ_{eff}^b (thin lines), and bolded lines are for ϵ_{eff} the isotropic arrangement.

(bold lines in Figure 4) are equal to the anisotropic case predictions for spheres ($a:b = 1$). For aspect ratios both greater than and less than 1, modeled permittivities migrate toward the permittivity value of the inclusion phase.

4.2. Three-Phase Modeling

Two confocal spheroids lying in a background phase (Figure 1) form the basis for modeling the more complex system of an unsaturated soil. There are six potential phase configurations (i.e., SWA, SAW, WSA, ASW, WAS, and AWS) which are presented by Friedman [1998] for spherical inclusions (the first letter designates the core or inner phase). Among these six configurations, Friedman [1998] found SWA and ASW to more closely describe the permittivity of natural unsaturated soils, where SWA tends to underestimate and ASW overestimates

the empirical predictions of Topp *et al.* [1980], but both maintain the solid-water interface characteristic of water-wetted minerals.

To model a partially wetted medium, we choose the solid phase (constant volume) as the reference ellipsoid from which to calculate the water and air phase volumes. For the SWA configuration, $V_0 = V_2/(1 - n)$, $V_1 = \theta V_0 + V_2$, and $V_2 = 4\pi a_2 b_2 c_2/3$, and for ASW, $V_0 = V_1/(1 - \theta)$, $V_1 = 4\pi a_1 b_1 c_1/3$, and $V_2 = V_0(n - \theta)$. We hold the solid phase core/shell as the reference volume, $V_{\text{ref}} = V_2$ for SWA and $V_{\text{ref}} = V_1$ for ASW. The solution for u_j shown in the Appendix A is given from the reference ellipsoid dimensions a , b , c , and the known water shell volume, V_1 for SWA, and from the air core volume, V_2 for ASW. Knowing the values of u_j , the depolarization factor of the ellipsoids N^i can be calculated and incorporated into (9)–(11) for finding the effective permittivity.

We compute permittivities of the SWA and ASW configurations for the ϵ_{eff}^a , ϵ_{eff}^b , and ϵ_{eff} scenarios using shapes of disks, spheres, and needles and plot these (Figure 5) as functions of volumetric water content θ for a porosity of 0.5. Similar to the two-phase mixtures, extreme aspect ratio disk-shaped inclusions act as thin layered sheets composed of the three phases, for which ϵ_{eff}^b approximates the arithmetic mean of the phase permittivities and ϵ_{eff}^a approximates the harmonic mean resulting for either phase configuration. As expected, for ASW at saturation and SWA at the dry end the permittivities converge to the two-phase SW and SA permittivities, respectively. However, for SWA at saturation and for ASW when dry, the background phases (dominant in the MG mixing) disappear which cause the resulting disparity in permittivities when compared to those resulting from the two-phase expression ((7) and (8) and $v = 0$). Long needle-shaped inclusions produce the arithmetic mean permittivity for ϵ_{eff}^a for either phase configuration, while for ϵ_{eff}^b , unique values result for either the SWA or ASW configuration owing to the two-dimensional circular nature of the needle scenario. The depolarization factors for spherical inclusions in any direction are of course the same, resulting in an isotropic medium. For the isotropic case (Figures 5g and 5h) the needle-shaped inclusion curve generally predicts permittivities greater than the Topp *et al.* [1980] equation. For the ASW configuration the computed permittivities for isotropic mixtures of spheres and needles are almost identical, while those of disk-shaped inclusions are much lower. Conversely, with SWA the sphere has the lowest predicted permittivity.

Among various methods of mixing the dual SWA-ASW system for best describing a moist soil, Friedman [1998] found the effective medium approximation (EMA), which treats each configuration symmetrically, to be an effective tool. The permittivity of each configuration is weighted according to the saturation degree S , a function of moisture content θ , and porosity n , as

$$f_w^{\text{SWA}} = 1 - S, \quad f_w^{\text{ASW}} = S, \quad S = \theta/n \quad (14)$$

with more weight given to SWA for low moisture while near saturation; ASW is more heavily weighted, resulting in the following solution of the parabolic two-component EMA equation [Friedman, 1998]

$$\epsilon_{\text{eff}} = \left(\frac{C^2}{16} + \frac{\epsilon_{\text{SWA}} \epsilon_{\text{ASW}}}{2} \right)^{1/2} - \frac{C}{4}, \quad (15)$$

where

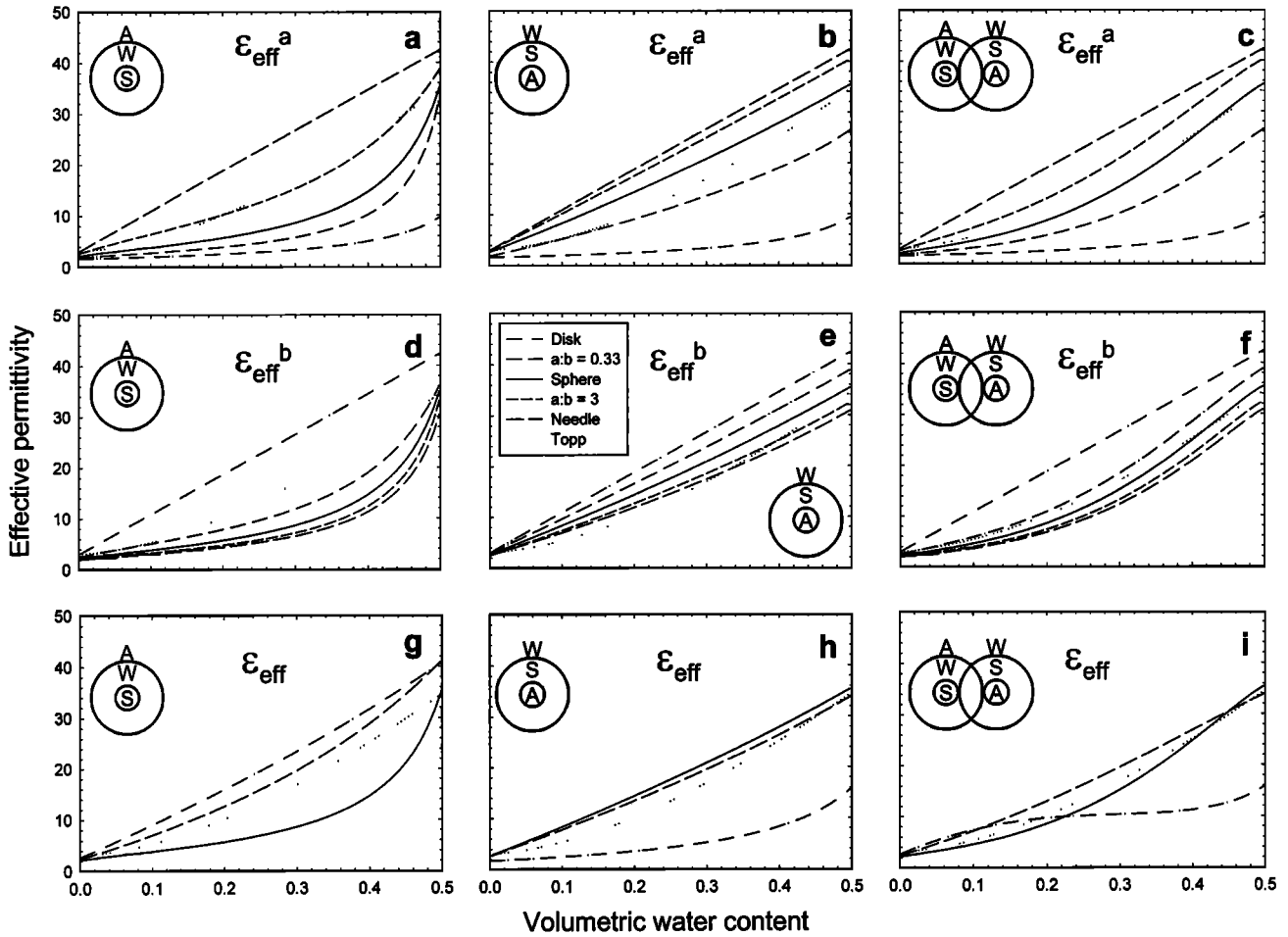


Figure 5. Effective permittivities as a function of water content and aspect ratio for the SWA and ASW three-phase configurations for (a and d) anisotropic cases ϵ_{eff}^a or (b and e) ϵ_{eff}^b and for (c and f) the isotropic case ϵ_{eff} . The saturation-degree weighted effective medium approximation (EMA) of the SWA and ASW configurations are given for (g) ϵ_{eff}^a or (h) ϵ_{eff}^b and for the isotropic case (i) ϵ_{eff} . Equations (9) and (10) were applied using a porosity of 0.5.

$$C = f_w^{\text{SWA}}(\epsilon_{\text{ASW}} - 2\epsilon_{\text{SWA}}) + f_w^{\text{ASW}}(\epsilon_{\text{SWA}} - 2\epsilon_{\text{ASW}}). \quad (16)$$

Results of the EMA approximation for ϵ_{eff}^a , ϵ_{eff}^b , and ϵ_{eff} are plotted in Figures 5c, 5f, and 5i. These results reflect averaging of the SWA and ASW configurations, while preserving particle shape effects. For spherical inclusions the mixing formulas of Friedman [1998] (for $n = 0.5$) are reconstructed in each case. The slight change in curvature from concave up to concave down, which is more pronounced for spherical particles (Figures 5c and 5f), and from concave down to concave up for isotropic disks (Figure 5i) is an artificial result of the EMA mixing of SWA and ASW configurations using the linear weight functions, (equation (14)). The curvature of the true $\epsilon_{\text{eff}}(\theta)$ relationships are expected to be fully concave up.

4.3. Two-Phase Measurements

TDR permittivity measurements in two-phase packings of water-saturated glass beads and in saturated mica particles are compared to MG predictions shown in Figure 6. Model predictions were computed using a glass bead permittivity of 7.5, a mica permittivity of 6.35, and a water permittivity of 78.5

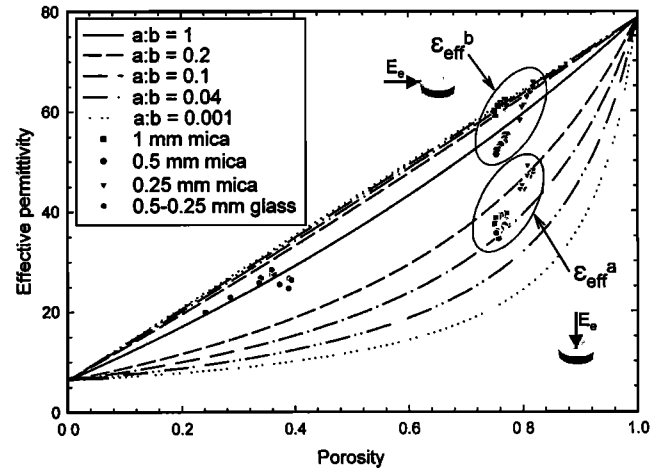


Figure 6. Effective dielectric permittivity measured in water-saturated mica and glass beads and modeled using the two-phase Maxwell-Garnett [1904] mixing theory for ϵ_{eff}^a and ϵ_{eff}^b . Particle aspect ratio, $a:b$, of mica is approximately 0.04 and as indicated for MG model predictions.

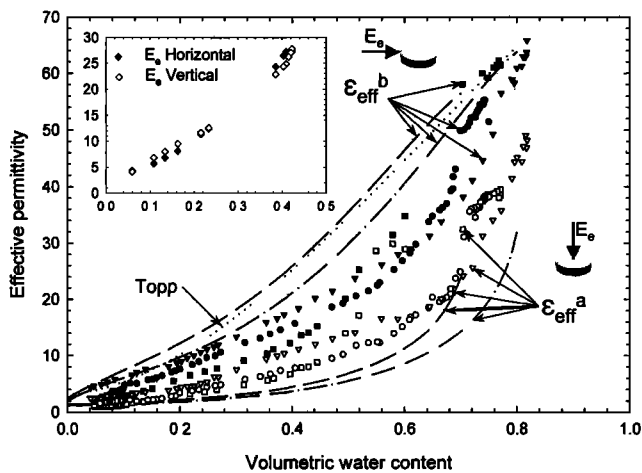


Figure 7. Effective permittivity of three different mica flake diameters plotted as a function of water content. Particle sizes of 0.25, 0.5, and 1 mm are represented by triangles, circles, and squares, respectively. Direction-dependent measurements are indicated by ϵ_{eff}^a (open symbols) and ϵ_{eff}^b (solid symbols). EMA model predictions for both electrical field alignments are shown for an aspect ratio of $a:b = 0.04$ and porosities of 0.7 (dashed line) and 0.8 (dashed-dotted line). Inset shows the effects of probe orientation on permittivity measurements in a packing of 0.5-mm glass beads using the same probe configuration as in the mica (i.e., one horizontal and one vertical probe).

(25°C). Monosized glass bead diameters varied from 0.025 to 0.5 mm, and to obtain lower porosities, binary mixtures of the same were also used. The predicted permittivity ($a:b = 1$) of the MG model is in agreement with measured ϵ_{eff} of the different glass bead packings. More interesting are the mica permittivity measurements which decrease because of the suction-induced shrinkage process, which reduces the porosity of the mica packing. The shrinkage is mostly vertical in direction and occurs prior to desaturation for the data shown in Figure 6. The mica measurements are generally correlated to the MG theory for ϵ_{eff}^b with an exception of the measurements in 0.5-mm mica, which lie significantly below the MG predictions. Modeled predictions of permittivity for ϵ_{eff}^a correspond to $a:b$ between 0.2 and 0.1, which are higher aspect ratios than the average $a:b$ of 0.04. Particle misalignment occurring near the rods and within the bedding plane is suspected as a deterrent to ideal particle- E_e field alignment. We expect partial disorder of this alignment to shift measurements to values between model predictions for ϵ_{eff}^a and ϵ_{eff}^b . Particle alignment is most critical near the central rod where the electrical field is the most concentrated (i.e., making the measurement more sensitive to the local conditions). In each of the three different particle size experiments, settling of from 7 to 13% of the initial mica column height occurred prior to air entry and subsequent desaturation. Some particle rearrangement is likely to occur near rod surfaces, depending on the amount of movement of the mica particles relative to the probes. Note also the slopes of the measured mica permittivity data in Figure 6, where, in particular, the measured ϵ_{eff}^b permittivities decrease relative to porosity at a faster rate than predicted by the MG model. Gaps developing around the probes during shrinkage may also influence readings, but according to Knight *et al.* [1997], a water-filled gap would only slightly increase the per-

mittivity reading. Some of the disparity may also arise from the circular nature of the E_e field resulting from a three-wire waveguide configuration, which for ϵ_{eff}^a produces a large spread in the resulting particle- E_e field alignment, which would artificially increase the resulting ϵ_{eff}^a according to (8), that is, as opposed to ϵ_{eff}^b whose E_e field runs fully parallel to the horizontal (bedding) plane of the mica. Salinity of the water phase in each medium may also have slightly influenced permittivity readings. However, interfacial polarization which occurs at low frequencies is not a likely factor owing to the relatively high effective frequency of the TDR device (1 GHz [Heimovarra *et al.*, 1996]). Maxwell-Wagner effects, which also occur at lower frequencies have been shown to increase permittivity readings, especially in the presence of platy particles [Hilfer, 1991; Kenyon, 1984; Sen, 1981]. Here we do not see evidence of increased permittivities and therefore suggest Maxwell-Wagner effects have a minimal influence on permittivity readings compared to the particle shape effects which are predicted by the dielectric mixing theory.

4.4. Three-Phase Measurements

To access the effect of particle shape on the effective permittivity of unsaturated media, we used the same mica flakes, which are approximately disk-shaped with an average aspect ratio of approximately 0.04. This aspect ratio provides a substantial difference in the depolarization factor as compared to that of a sphere (Figure 2), and the differences in permittivities parallel and perpendicular to the bedding plane are expected to be significant based on dielectric mixing theory. Permittivity measurements were made using three-wire TDR probes, positioned in order to orient electrical field lines between the inner and outer rods perpendicular (ϵ_{eff}^a) and parallel (ϵ_{eff}^b) to the bedding plane of the mica flakes (Figure 3). For testing the experimental setup we also measured the ϵ_{eff}^a and ϵ_{eff}^b of glass beads, which are expected to be the same for the entire range of volumetric water content. Figure 7 (inset) shows the minimal impact which probe orientation has on horizontal and vertical permittivity measurements in 0.5-mm glass beads. The observed small differences in horizontal and vertical measurements are likely a result of hydrostatic water distribution and its effect on the propagation of electromagnetic waves along the waveguides.

Measured permittivities in three differently size fractions of mica particles are generally correlated to each other and to the electrical field alignment as determined by probe positioning (Figure 7). Substantial deviations in permittivity data from the generally observed trend are attributable to the hydrostatic distribution of lower moisture contents in the upper part of the probe's volume of influence which is highly correlated to particle size (i.e., most significant for ϵ_{eff}^a of 1-mm particles). In the permittivity calculations we used the average aspect ratio of 0.04, two different porosities of 0.7 to 0.8, and dielectric constants of 6.35 for mica, 78.5 for water, and 1 for air. Measured permittivity data generally lie between the ϵ_{eff}^a and ϵ_{eff}^b model predictions. As discussed in section 4.3, we anticipate a general gravitation of permittivity readings toward a central (isotropic) value between the predicted permittivities for the two principal directions because of particle-field misalignment. Measured data of both ϵ_{eff}^a and ϵ_{eff}^b lie well below the Topp *et al.* [1980] curve except for ϵ_{eff}^b values near saturation, while MG-based model predictions for ϵ_{eff}^b are near the Topp curve.

Assumptions and limitations of the proposed mixing models should be mentioned. The use of a single aspect ratio ellipsoi-

dal geometry to model soil particles is of limited justification. Additionally, the elliptical water shells are not consistent with the water-air interface behavior except for very low moisture contents. Modeling the phases of solid, water, and air using a confocal arrangement, while mathematically efficient, does not ideally mimic the complex and variable configuration of partially saturated soils. Some of these discrepancies are partially corrected by the EMA mixing of the SWA and ASW phase configurations.

Summarizing, expressions for two- and three-phase dielectric systems give predictions of the effective permittivities which are relevant to soils and other porous media having uniformly shaped particles. The Maxwell-Garnett form of these expressions has the advantage of being written explicitly for ϵ_{eff} . Since many disk-like media are characteristically high in surface area, resulting in bound water characterized by reduced permittivities, we have adopted a method of incorporating bound water effects into the permittivity calculation.

4.5. Combined Effects of Bound Water and Particle Shape

In a "free" state, water molecules exhibit relatively free rotation (inhibited only by neighboring molecules) in the presence of electromagnetic fields, but when these molecules make up the vicinal layers near solid surfaces, their mobility is reduced by the interaction with solid molecules, and the resulting dielectric permittivity is much lower than that of free water [Mulla *et al.*, 1984; Or and Wraith, 1999]. Bound forms of water have been found to possess lower dielectric constants depending on their degree of mobility, which has been shown to be a function of their distance from the solid surface [Kurosaki, 1954; Thorp, 1959]. In order to account for the reduced permittivity of bound water in our mixing scheme, we use the water film-thickness dependent approach taken by Friedman [1998], where an exponential function is used for describing the water phase permittivity starting from a minimum ($\epsilon_{\text{min}} = 5.5$) near the mineral surfaces toward a maximum (ϵ_{max}) permittivity, which describes the free water phase far away from the solid walls, resulting in

$$\epsilon_w = d_w \epsilon_{\text{max}} / d_w + \frac{1}{\lambda} \ln \left[\frac{\epsilon_{\text{max}} - (\epsilon_{\text{max}} - \epsilon_{\text{min}}) e^{-\lambda d_w}}{\epsilon_{\text{min}}} \right] \quad (17)$$

for the average permittivity of the water phase, where λ is an empirical decay factor taken to be 10^8 cm^{-1} and d_w is the approximate thickness of the water film spread uniformly over the surface of the solid inclusion. The water film thickness is taken for a single inclusion rather than the bulk media as given by Friedman [1998] and is written here as

$$d_w = \theta V_s / S_A (1 - n), \quad (18)$$

where θ is the volumetric water content, V_s is the entire volume contained within the shell of the solid-phase ellipsoid (includes air phase for ASW), S_A is the ellipsoid surface area, and n is the porosity. Equation (17) yields a dimensionless $\epsilon_w(a:b, a\lambda, \theta, n)$ relationship which we will apply to a single spheroid of unit volume ($V_{\text{rep}} = 1$). The surface area of an oblate (disk like) ellipsoid may be computed based on its eccentricity e and axial dimensions as [Hodgman *et al.*, 1954, p. 317]

$$S_A = 2\pi b^2 + \frac{\pi a^2}{e} \ln \left(\frac{1+e}{1-e} \right), \quad e = \frac{\sqrt{b^2 - a^2}}{b}, \quad (19)$$

and the surface area for a prolate (needle like) ellipsoid is

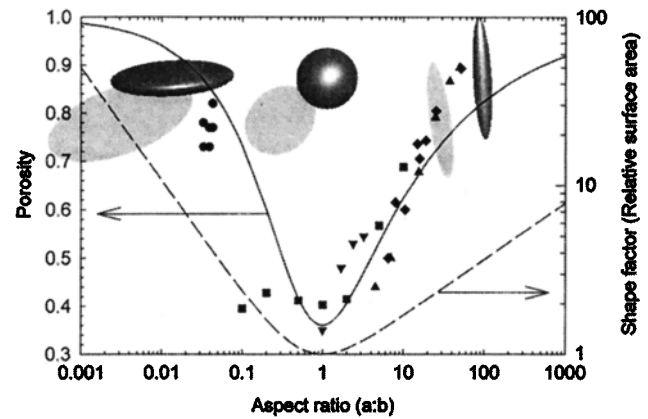


Figure 8. Spheroid porosity and shape factor (spheroid surface area per surface area of a sphere of equal volume) as a function of aspect ratio. Data shown are for mica (circles) (this study), silica-hematite prolates (inverted triangles) [Thies-Weesie *et al.*, 1995], numerical data (squares) [Coehlo *et al.*, 1997], wooden rods (triangles) and fiberglass (diamonds) [Milewski, 1978].

$$S_A = 2\pi b^2 + \frac{2\pi ab}{e} \sin^{-1} e, \quad e = \frac{\sqrt{a^2 - b^2}}{a}. \quad (20)$$

By dividing the spheroid surface area by the surface area of a sphere of unit volume (i.e., V_{rep}), we obtain the shape factor S_F , which is plotted as a function of aspect ratio in Figure 8. A power law relationship for both needle-shaped and disk-shaped ellipsoids (exception occurring near the sphere transition) is shown, the disk-like ellipsoids exhibiting larger relative surface areas for given aspect ratios compared to needle-shaped ellipsoids. The shape factor has been shown to be related to the porosity of randomly arranged particles of nonspherical shape. Using an empirical correlation, the porosity n can be expressed as a function of aspect ratio through surface area and volume as [Cumberland and Crawford, 1987]

$$n = 1 - k \frac{V_{\text{rep}}^{2/3}}{S_A}, \quad (21)$$

where S_A is determined from (19) or (20) and k is an empirical constant which can be used to scale the resulting curve to a reference point. In this case we use a value of $k = 3.10$ [Thies-Weesie *et al.*, 1995] which scales the curve to the porosity of randomly packed monosized spheres ($n = 0.36$) at an aspect ratio of 1 shown in Figure 8. Data from our study and other porosity measurements of ellipsoids as well as numerical data are compared to (21) in Figure 8. Porosities of nonspherical particles shown are those resulting from deposition under the influence of gravity, and therefore particles are more likely to be aligned with the bedding plane than to be randomly arranged. The curve from (21), representing random orientation based on spheres, is expected to overpredict the porosity of partially aligned ellipsoids, as it does for the disk-shaped particle data shown, but for needle-shaped particles the empirical expression underpredicts porosity when compared to data of various ellipsoidally and cylindrically shaped particles. The general shape of the curve, however, represents the general trend of porosity to increase as aspect ratio deviates from 1. The potential range of porosities for any given aspect ratio can vary greatly and is largely a function of the particle orien-

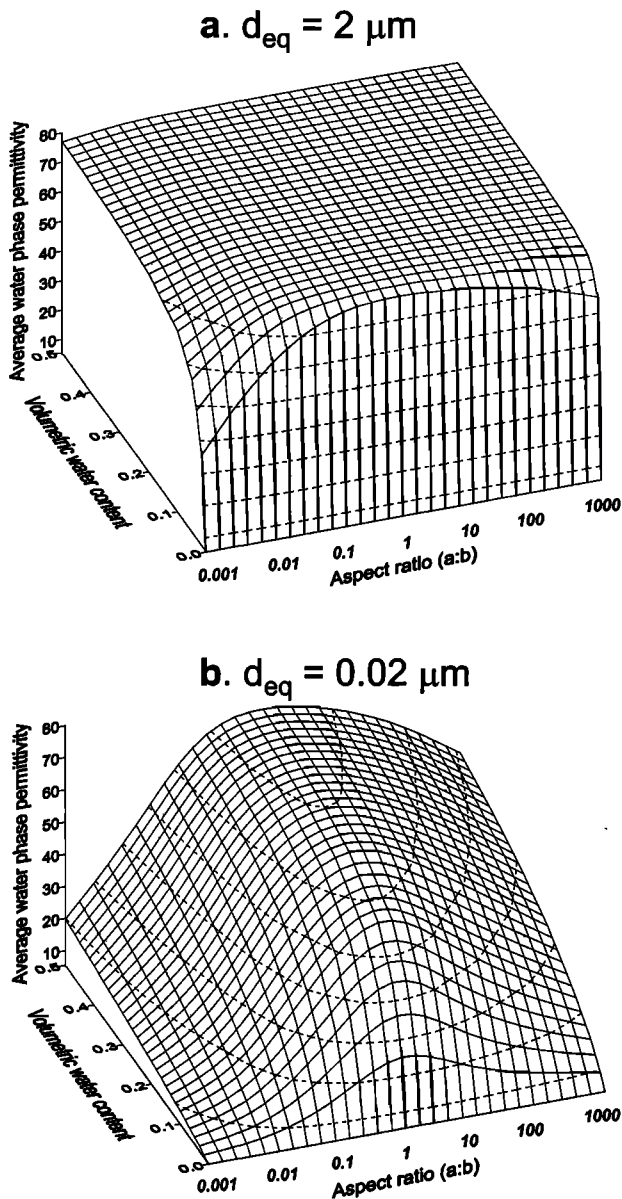


Figure 9. Average water phase permittivity ϵ_w as a function of water content and particle aspect ratio for particles whose equivalent diameter is (a) $2 \mu m$ and (b) $0.02 \mu m$.

tation imposed by the force of gravity during packing, which for extreme aspect ratio particles tends to align the larger axis of disks and needles perpendicular to the force of gravity. Higher porosities where particles are more disperse and free-floating may occur in equal buoyancy solid-fluid systems or in the microgravity environment of space [Jones and Or, 1999].

For multiple confocal ellipsoids the distance between an inner and outer shell at each semiaxis varies according to the aspect ratio (Figure 1), where for a sphere the distance is the same along each axis, a , b , or c . For nonspheres the minimum water film thickness occurs along the rotation axis for needle-like spheroids and along the common axes for disk-like spheroids where this distance is computed as $a_1 - a_2$ and $b_1 - b_2$ for the rotation and common axes, respectively. For estimating bound water effects on the effective permittivity, we are interested in the water film thickness d_w , which surrounds the solid phase, and in order to quantify its dependence on aspect ratio,

we assume some average of the minimum and maximum thicknesses. For thin-disk and long-needle shapes the minimum and maximum d_w in each case differ by as much as 3 orders of magnitude. Since the maximum film thickness is associated with the greatest portion of the surface area (e.g., face of a disk and length of a needle), we simplify the calculation of the average d_w by assuming the water is spread uniformly over a flat solid surface of the same surface area.

On the basis of surface area of a single spheroid the water phase permittivities as a function of aspect ratio ($10^{-3} \leq a:b \leq 10^3$) and water content ($0 \leq \theta \leq 0.5$) for an equivalent spherical particle diameter d_{eq} of $2 \mu m$ and $0.02 \mu m$ were computed and plotted in Figure 9. The reduction in the water phase permittivity for $d_{eq} = 2 \mu m$ (upper limit of clay size) is negligible for spherical particles, while disk-shaped particles ($a:b = 10^{-3}$) show a slight reduction at low water contents. For $d_{eq} = 0.02 \mu m$ particle both disk- and needle-shaped particles have larger reductions than spherical particles throughout the entire range of water contents. The shape effect on surface area (and thus bound water) is very evident for $d_{eq} = 0.02 \mu m$ showing disk-shaped particles having saturated water phase permittivities of only 20, while saturated spherical particles have only a slight reduction from 80 to about 75.

Introduction of the surface area-dependent water phase permittivity into the mixing theory enables the analysis of the combined particle shape and bound water effects on the dielectric permittivity of monosized particle mixtures. From (17) the water phase permittivity is determined, and the resulting effective permittivity of the mixture is computed based on the EMA mixing of the SWA-ASW phase configurations. The resulting permittivity predictions of the EMA for spherical inclusions reproduces the results of Friedman [1998] for a given surface area and porosity. The resulting permittivities for disks, spheres, and needles are only slightly different than shown in Figure 5i (i.e., no bound water) for a particle size of $2 \mu m$ (Figure 10a), where the effects of bound water are small. The bound water effect is substantially greater for $d_{eq} = 0.02 \mu m$ as seen in Figure 10b, where extreme aspect ratio inclusions show greatly reduced effective permittivities compared to only a small reduction in the case of the spherical inclusion curve. Effective permittivity curves for aspect ratios of 0.1 and 10 give a sense of the effects of intermediate values, with the needle ($a:b = 10^3$) inclusion curve being comparable to a disk of aspect ratio 0.1 for the case $d_{eq} = 0.02 \mu m$. On the basis of three-phase dielectric mixing theory coupled with surface area-dependent estimates of bound water, particle shape effects play a substantial role in altering the effective permittivity of porous media. Measured permittivities in materials containing highly oblate or prolate particles likely are influenced by particle shape effects, which for submicrometer-sized particles are also influenced by the reduced permittivities of bound water.

Summarizing, the surface area of a rotation ellipsoid was used to determine the average thickness of water surrounding the solid phase inclusion assuming an equivalent flat surface. The film thickness allowed calculation of the water phase permittivity accounting for bound water, which has the effect of lowering the effective permittivity below that which results from shape effects alone.

5. Conclusion

Particle shape effects on the effective permittivity of particulate mixtures were mathematically analyzed using dielectric

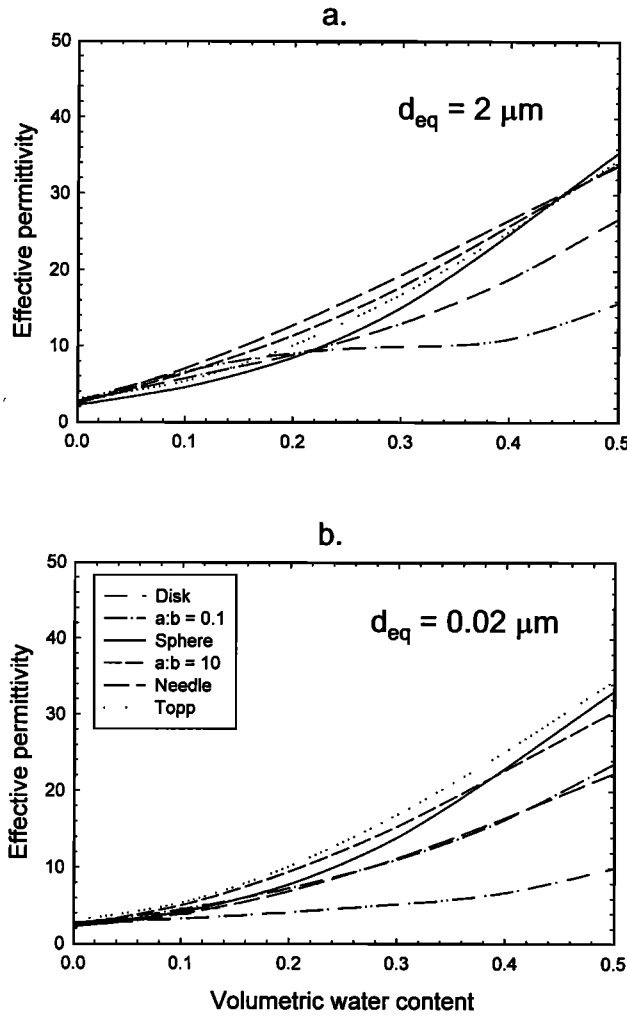


Figure 10. Effective permittivity predictions for isotropic media using an EMA of three-phase mixtures of SWA and ASW as a function of volumetric water content. Predictions incorporate the reduced average water phase permittivity due to bound water for particles of equivalent diameter of (a) $2 \mu\text{m}$ and (b) $0.02 \mu\text{m}$.

mixing theories. Two- and three-phase systems were modeled using ellipsoids of revolution as geometrical descriptors. For the three-phase system composed of confocal inclusions, a symmetric effective medium approximation of two different phase configurations (ASW and SWA) was used to approximate both anisotropic and isotropic unsaturated porous media. Disk- and needle-shaped particles were shown to alter the effective permittivity of the mixture compared to a spherical shape by shifting the effective permittivity to a value closer to that of the inclusions. TDR permittivity measurements in unsaturated anisotropic packings of three different sizes of mica particles were well below predictions of the *Topp et al.* [1980] equation. In isotropic soils containing significant amounts of disk-shaped particles (i.e., clayey soils), shape effects could play a significant role in the reduced permittivities which are typically attributed to bound forms of water. This is likely to occur at higher water contents where water films are relatively thick and relaxation interactions are few. The extent of the particle shape effect on permittivity is of the order of magnitude of the variation in permittivity due to porosity (density)

and bound water effects. Particle shape effects were demonstrated to be a significant factor affecting dielectric permittivity measurements and become most significant where particle aspect ratio deviates by more than an order of magnitude from the aspect ratio of a sphere.

Appendix A

For the case of a general ellipsoid, (2) was solved using Mathematica [Wolfram Research, Inc., 1996] producing an explicit expression for u_j as a function of the known ellipsoid volume V_j (i.e., V_1 for SWA and V_2 for ASW configurations) and axial radii a , b , and c of the reference ellipsoid, given as

$$u_j = -\frac{f_1}{3} - \left[\frac{f_4}{(2^{2/3})24\pi^2(f_5 + \sqrt{4f_4^3 + f_5^2})^{1/3}} \right] + \left[\frac{(f_5 + \sqrt{4f_4^3 + f_5^2})^{1/3}}{(2^{1/3})48\pi^2} \right],$$

where

$$\begin{aligned} f_1 &= a^2 + b^2 + c^2, \\ f_2 &= (ab)^2 + (ac)^2 + (cb)^2, \\ f_3 &= 6912\pi^4(16(abc)^2\pi^2 - 9V_j^2), \\ f_4 &= -256f_1^2\pi^4 + 768f_2\pi^4, \\ f_5 &= -8192f_1^3\pi^6 + 36864f_1f_2\pi^6 - f_3. \end{aligned}$$

Notation

- a rotation axis of spheroid, radial dimension, L .
- b, c common axes of spheroid, radial dimension, L .
- d_{eq} equivalent diameter of a sphere having equal volume to the spheroid, L .
- d_w water film thickness on the surface of an inclusion, L .
- e eccentricity of inclusion.
- k empirical constant.
- D electrical flux density, $Q L^{-2}$.
- E_e electrical field external to ellipsoid, $M L T^{-2} Q^{-1}$.
- f_w weight function for EMA mixing.
- L length.
- M mass.
- N' ellipsoid depolarization factor where E_e runs parallel to the i th axis.
- n porosity.
- n_v number of inclusions per unit volume, L^{-3} .
- P polarization, $Q L^{-2}$.
- p dipole moment, $Q L$.
- Q electrical charge.
- S saturation degree, equal to θ/n .
- S_A surface area of a spheroid, L^2 .
- T time.
- u_j scalar used to size external/internal ellipsoid dimensions relative to $u = 0$, L^2 .
- V_j ellipsoid volume of the j th phase (see Figure 1), L^3 .
- V_s volume contained within the outer shell of the solid phase, L^3 .
- V_{rep} representative volume equal to $\pi d_{eq}^3/6$, L^3 .
- V_t total representative volume or unit cell, L^3 .
- α polarizability, $Q^2 T^2 M^{-1}$.

- ϵ_{eff}^a effective mixture permittivity where E_e is parallel to "a."
 ϵ_{eff}^b effective mixture permittivity where E_e is parallel to "b."
 ϵ_{eff} effective isotropic mixture permittivity.
 ϵ_{max} maximum water-phase permittivity.
 ϵ_{min} minimum water-phase permittivity.
 ϵ_0 permittivity of background phase.
 ϵ_1 permittivity of phase lying between core inclusion and background.
 ϵ_2 permittivity of core inclusion phase.
 ϵ_w permittivity of water.
 λ empirical decay factor, L^{-1} .
 θ volumetric water content.
 ϕ_j volumetric fraction of the j th phase (i.e., $\phi_1 = V_1/V_t$).

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